GATE CS 2026 to 2027: Key Syllabus Changes

A clean summary based on the official syllabus screenshots supplied

Subject	Change	Status
General Aptitude	No content change.	UNCHANGED
Engineering Mathematics	No content change.	UNCHANGED
Digital Logic	Minimization methods and circuit-design wording are expanded.	EXPANDED
Computer Organization	ALU/control-unit design and cache mapping/performance are explicitly added.	ADDED
Programming, Algorithms, TOC, Compiler and OS	No substantive content change.	UNCHANGED
Databases	No content change.	UNCHANGED
Computer Networks	Major wording and scope revision; several 2026 items are not explicitly listed.	NOT LISTED

'Not explicitly listed' means the topic is absent from the visible GATE 2027 wording; it does not necessarily mean the concept can never be tested.

GATE 2026 vs GATE 2027 - General Aptitude

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GA	General Aptitude

Verbal Aptitude

Basic English grammar: tenses, articles, adjectives, prepositions, conjunctions, verb-noun agreement, and other parts of speech

Basic vocabulary: words, idioms, and phrases in context.

Reading and comprehension, Narrative sequencing.

Quantitative Aptitude

Data interpretation: data graphs (bar graphs, pie charts, and other graphs representing data), 2- and 3-dimensional plots, maps, and tables

Numerical computation and estimation: ratios, percentages, powers, exponents and logarithms, permutations and combinations, and series Mensuration and geometry Elementary statistics and probability.

Analytical Aptitude

Logic: deduction and induction, Analogy, Numerical relations and reasoning.

Spatial Aptitude

Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping paper folding, cutting, and patterns in 2 and 3 dimensions.

GATE 2027 — IIT Madras - Organizing Institute

GA	General Aptitude

Section 1:	Verbal Aptitude
Basic English grammar: tenses, articles, adjectives, prepositions, conjunctions, verb-noun agreement, and other parts of speech. Basic vocabulary: words, idioms, and phrases in context. Reading and comprehension, Narrative sequencing.	
Section 2:	**Quantitative Aptitude**
Data interpretation: data graphs (bar graphs, pie charts, and other graphs representing data), 2-and 3-dimensional plots, maps, and tables. Numerical computation and estimation: ratios, percentages, powers, exponents and logarithms, permutations and combinations, and series. Mensuration and geometry, Elementary statistics and probability.	
Section 3:	**Analytical Aptitude**
Logic: deduction and induction, Analogy, Numerical relations and reasoning.	
Section 4:	**Spatial Aptitude**
Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping paper folding, cutting, and patterns in 2 and 3 dimensions.	

Highlighted changes from GATE 2026 to GATE 2027

UNCHANGED The General Aptitude syllabus content is the same in GATE 2026 and GATE 2027.

FORMAT GATE 2027 organizes the same content into four numbered sections.

GATE 2026 vs GATE 2027 - Engineering Mathematics and Digital Logic

GATE 2026

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CS — Computer Science and Information Technology

Section 1: Engineering Mathematics

Discrete Mathematics: Propositional and first order logic. Sets, relations, functions, partial orders and lattices. Monoids, Groups. Graphs: connectivity, matching, colouring. Combinatorics: counting, recurrence relations, generating functions.

Linear Algebra: Matrices, determinants, system of linear equations, eigenvalues and eigenvectors, LU decomposition.

Calculus: Limits, continuity and differentiability, Maxima and minima, Mean value theorem, Integration.

Probability and Statistics: Random variables, Uniform, normal, exponential, Poisson and binomial distributions. Mean, median, mode and standard deviation. Conditional probability and Bayes theorem.

Section 2: Digital Logic

Boolean algebra. Combinational and sequential circuits. Minimization. Number representations and computer arithmetic (fixed and floating point).

GATE 2027

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CS — Computer Science and Information Technology

Section 1: Engineering Mathematics

Discrete Mathematics: Propositional and first order logic. Sets, relations, functions, partial orders and lattices. Monoids, Groups. Graphs: connectivity, matching, colouring. Combinatorics: counting, recurrence relations, generating functions.

Linear Algebra: Matrices, determinants, system of linear equations, eigenvalues and eigenvectors, LU decomposition.

Calculus: Limits, continuity and differentiability, Maxima and minima, Mean value theorem, Integration.

Probability and Statistics: Random variables, Uniform, normal, exponential, Poisson and binomial distributions. Mean, median, mode and standard deviation. Conditional probability and Bayes theorem.

Section 2: Digital Logic

Boolean algebra and minimization – algebraic technique, Karnaugh map, tabular method. Design of combinational and sequential circuits. Number representation and arithmetic (fixed and floating point).

Highlighted changes from GATE 2026 to GATE 2027

- **UNCHANGED** Engineering Mathematics remains unchanged.
- **ADDED** Digital Logic explicitly lists minimization methods: algebraic technique, Karnaugh map and tabular method.
- **EXPANDED** The 2027 wording explicitly emphasizes the design of combinational and sequential circuits.
- **WORDING** Number representations and computer arithmetic is shortened to number representation and arithmetic.

GATE 2026 vs GATE 2027 - Computer Science Sections 3-8

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Section 3: Computer Organization and Architecture

Machine instructions and addressing modes. ALU, data path and control unit. Instruction pipelining, pipeline hazards. Memory hierarchy: cache, main memory and secondary storage; I/O interface (interrupt and DMA mode).

Section 4: Programming and Data Structures

Programming in C. Recursion. Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs.

Section 5: Algorithms

Searching, sorting, hashing. Asymptotic worst case time and space complexity. Algorithm design techniques: greedy, dynamic programming and divide and conquer. Graph traversals, minimum spanning trees, shortest paths.

Section 6: Theory of Computation

Regular expressions and finite automata. Context-free grammars and push-down automata. Regular and context-free languages, pumping lemma. Turing machines and undecidability.

Section 7: Compiler Design

Lexical analysis, parsing, syntax-directed translation. Runtime environments. Intermediate code generation. Local optimisation, Data flow analyses: constant propagation, liveness analysis, common sub expression elimination.

Section 8: Operating System

System calls, processes, threads, inter process communication, concurrency and synchronization. Deadlock. CPU and I/O scheduling. Memory management and virtual memory. File systems.

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Section 3:	Computer Organization and Architecture
	Instruction set and addressing modes. Design of arithmetic and logic unit (ALU). Design of control unit – hardwired and microprogrammed. Memory interfacing and hierarchy: performance, cache memory mapping. I/O interface (interrupt and DMA). Instruction pipelining, pipeline hazards.
Section 4:	Programming and Data Structures
	Programming in C. Recursion. Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs.
Section 5:	Algorithms
	Searching, sorting, hashing. Asymptotic worst case time and space complexity. Algorithm design techniques: greedy, dynamic programming and divide-and-conquer. Graph traversals, minimum spanning trees, shortest paths.
Section 6:	Theory of Computation
	Regular expressions and finite automata. Context-free grammars and push-down automata. Regular and context-free languages, pumping lemma. Turing machines and undecidability.
Section 7:	Compiler Design
	Lexical analysis, parsing, syntax-directed translation. Runtime environments. Intermediate code generation. Local optimisation, Data flow analyses: constant propagation, liveness analysis, common sub expression elimination.
Section 8:	Operating System
	System calls, processes, threads, inter-process communication, concurrency and synchronization. Deadlock. CPU and I/O scheduling. Memory management and virtual memory. File systems.

Highlighted changes from GATE 2026 to GATE 2027

- **ADDED** Computer Organization explicitly adds ALU design and control-unit design, including hardwired and microprogrammed control.
- **ADDED** Memory coverage now explicitly includes performance and cache-memory mapping.
- **WORDING** The 2026 wording on cache, main memory and secondary storage is replaced by memory interfacing and hierarchy.
- **UNCHANGED** Programming, Algorithms, Theory of Computation, Compiler Design and Operating System are substantively unchanged.

GATE 2026 vs GATE 2027 - Databases and Computer Networks

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Section 9: Databases

ER?model. Relational model: relational algebra, tuple calculus, SQL. Integrity constraints, normal forms. File organization, indexing (e.g., B and B+ trees). Transactions and concurrency control.

Section 10: Computer Networks

Concept of layering: OSI and TCP/IP Protocol Stacks; Basics of packet, circuit and virtual circuit- switching; Data link layer: framing, error detection, Medium Access Control, Ethernet bridging; Routing protocols: shortest path, flooding, distance vector and link state routing; Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT); Transport layer: flow control and congestion control, UDP, TCP, sockets; Application layer protocols: DNS, SMTP, HTTP, FTP, Email.

GATE 2027 — IIT Madras - Organizing Institute

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Section 9:	Databases
ER-model. Relational model: relational algebra, tuple calculus, SQL. Integrity constraints, normal forms. File organization, indexing (e.g., B and B+ trees). Transactions and concurrency control.	
Section 10:	**Computer Networks**
Principles of Layering; Basics of switching (circuit, packet and virtual circuit) and performance metrics; Data link layer: error detection, Medium Access Control, Ethernet; Distance vector and link state routing; IPv4 - Fragmentation, CIDR Notation, Network Address Translation; TCP- flow control and congestion control, socket API; DNS and HTTP.	

Highlighted changes from GATE 2026 to GATE 2027

- **UNCHANGED** Databases remains unchanged.
- **ADDED** Computer Networks explicitly adds performance metrics and uses the term socket API.
- **NOT LISTED** OSI and TCP/IP protocol stacks, framing, Ethernet bridging, shortest-path routing and flooding are not explicitly listed in the 2027 wording.
- **NOT LISTED** ARP, DHCP, ICMP, UDP, SMTP, FTP and Email are not explicitly listed in the 2027 wording.
- **RETAINED** DNS, HTTP, TCP flow and congestion control, IPv4 fragmentation, CIDR and NAT remain.